

BREMERTON, WA, USAWMO: **727928**Lat: **47.483N**Lon: **122.767W**Elev: **135**StdP: **99.71**Time Zone: **-8.00 (NAP)**Period: **94-19**WBAN: **94263****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-5.1	-2.9	-11.0	1.5	-1.6	-7.8	2.0	-0.7	10.7	7.2	9.1	6.8	1.3	40	0.552

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	30.2	18.7	27.9	17.7	26.2	17.0	19.4	28.6	18.4	26.9	17.5	25.0	3.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.1	11.6	22.0	14.9	10.7	20.8	14.0	10.1	19.8	55.4	28.8	52.3	26.9	49.3	25.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
							Mean		Standard Deviation		n=5 years		n=10 years	
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.8	7.7	6.8	-8.5	34.0	2.9	2.3	-10.6	35.7	-12.3	37.0	-13.9	38.3	-16.0	40.0
			-9.0	20.8	2.6	1.1	-10.9	21.6	-12.5	22.2	-14.0	22.8	-15.9	23.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	4.1	4.6	6.5	8.6	12.0	14.7	17.6	17.7	15.0	10.0	6.2	3.4	(6)
(7)		DBStd	5.82	3.17	2.86	2.88	3.05	3.13	2.94	2.79	2.49	2.75	2.72	3.09	3.17	(7)
(8)		HDD10.0	883	184	151	114	62	13	1	0	0	1	34	119	204	(8)
(9)		HDD18.3	3079	442	384	367	292	197	118	47	42	106	259	364	462	(9)
(10)		CDD10.0	908	1	1	6	20	76	141	236	238	149	34	5	1	(10)
(11)		CDD18.3	61	0	0	0	0	1	8	24	22	5	0	0	0	(11)
(12)		CDH23.3	1147	0	0	2	11	47	150	434	386	117	1	0	0	(12)
(13)		CDH26.7	346	0	0	0	1	7	47	151	115	26	0	0	0	(13)
(14)	Wind	WSAvg	2.5	2.6	2.6	3.0	2.7	2.5	2.5	2.3	2.2	2.3	2.4	2.6	2.5	(14)
(15)	Precipitation	PrecAvg	1342	207	141	147	94	58	42	18	29	47	133	214	211	(15)
(16)		PrecMax	1691	426	367	310	216	117	86	52	93	203	313	506	370	(16)
(17)		PrecMin	996	97	5	36	31	5	6	3	1	3	16	60	71	(17)
(18)		PrecStd	220	74	71	61	49	31	22	14	26	44	71	101	84	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.8	14.1	20.5	24.8	27.5	31.4	33.9	32.3	29.1	22.2	15.6	12.3	(19)
(20)			MCWB	10.4	8.6	11.9	14.3	16.2	18.5	20.4	19.1	17.7	14.2	12.8	11.2	(20)
(21)		2%	DB	11.2	12.3	16.2	20.1	24.1	27.7	30.9	29.4	27.0	18.7	13.0	10.9	(21)
(22)			MCWB	9.5	8.6	10.7	12.2	15.5	17.4	19.2	18.6	16.8	12.8	11.2	9.5	(22)
(23)		5%	DB	9.8	11.0	13.0	17.3	22.0	24.1	27.7	27.6	23.9	16.6	12.4	9.0	(23)
(24)			MCWB	8.4	8.4	8.9	10.9	14.3	16.0	18.0	17.8	15.9	12.3	10.9	7.9	(24)
(25)		10%	DB	8.7	9.0	11.5	14.2	18.8	21.9	25.8	25.3	22.0	14.8	11.2	7.8	(25)
(26)			MCWB	7.6	7.2	8.3	9.2	13.0	14.9	17.1	16.9	15.0	11.5	9.8	6.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.8	11.4	13.9	15.3	17.4	19.7	21.0	20.3	19.1	16.2	13.6	11.6	(27)
(28)			MCD B	12.3	12.1	16.3	22.6	24.8	29.4	31.9	29.5	27.0	18.8	14.8	12.0	(28)
(29)		2%	WB	9.8	10.0	11.4	12.9	16.0	17.7	19.6	19.1	17.7	14.3	12.2	9.3	(29)
(30)			MCD B	10.6	11.3	14.8	18.9	23.1	26.3	29.3	27.9	24.6	16.8	13.0	10.1	(30)
(31)		5%	WB	8.4	8.4	9.9	11.5	14.7	16.4	18.4	18.3	16.6	13.1	11.1	8.0	(31)
(32)			MCD B	9.1	9.5	12.7	16.0	20.8	23.8	27.2	26.5	22.7	15.7	12.0	8.5	(32)
(33)		10%	WB	7.6	7.6	8.6	10.2	13.3	15.1	17.3	17.3	15.4	12.0	9.8	7.1	(33)
(34)			MCD B	8.3	8.6	10.9	13.9	18.3	20.9	25.0	24.6	20.0	14.0	11.0	7.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	7.5	8.7	10.2	11.0	11.1	13.1	13.1	11.7	8.9	6.8	5.7	(35)
(36)			MCD BR	7.2	10.4	12.1	15.0	16.7	16.9	18.1	18.0	16.8	12.0	8.2	6.4	(36)
(37)			MCWBR	5.5	7.2	7.4	8.6	8.9	8.6	8.7	8.8	8.8	7.6	6.1	5.3	(37)
(38)		5% WB	MCD BR	5.7	7.6	10.7	13.1	14.6	15.4	17.0	16.5	14.5	10.1	6.0	5.4	(38)
(39)			MCWBR	4.9	6.0	7.1	7.8	8.1	8.2	8.5	8.3	7.9	7.0	5.1	4.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.324	0.311	0.324	0.345	0.370	0.356	0.344	0.348	0.336	0.325	0.322	0.324	(40)	
(41)		taud	2.494	2.570	2.544	2.460	2.408	2.470	2.529	2.519	2.548	2.577	2.527	2.435	(41)	
(42)		Ebn at Noon	761	867	904	910	894	907	914	897	878	836	758	704	(42)	
(43)		Edh at Noon	65	73	86	103	113	107	99	96	85	71	62	61	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.00	1.86	2.80	4.09	5.18	5.66	6.32	5.40	3.94	2.21	1.17	0.82	(44)	
(45)		RadStd	0.14	0.25	0.33	0.39	0.42	0.59	0.49	0.36	0.37	0.24	0.15	0.12	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (61 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page